Machine Learning 2026 - Assignment 2

Practical clustering experiments | Roll number: 2024514

1. Palmer Penguins: K-Means and Gaussian Mixture Models

1.1 Exploratory data analysis and preprocessing

Item	Result
Observations	344
Selected numerical features	4
Features used	bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g
Rows with at least one missing selected measurement	2
Imputation method	Median imputation by feature

Feature	Missing	Median used	Scaled mean	Scaled std
bill_length_mm	2	44.45	0.0000	1.0000
bill_depth_mm	2	17.30	0.0000	1.0000
flipper_length_mm	2	197.00	0.0000	1.0000
body_mass_g	2	4050.00	0.0000	1.0000

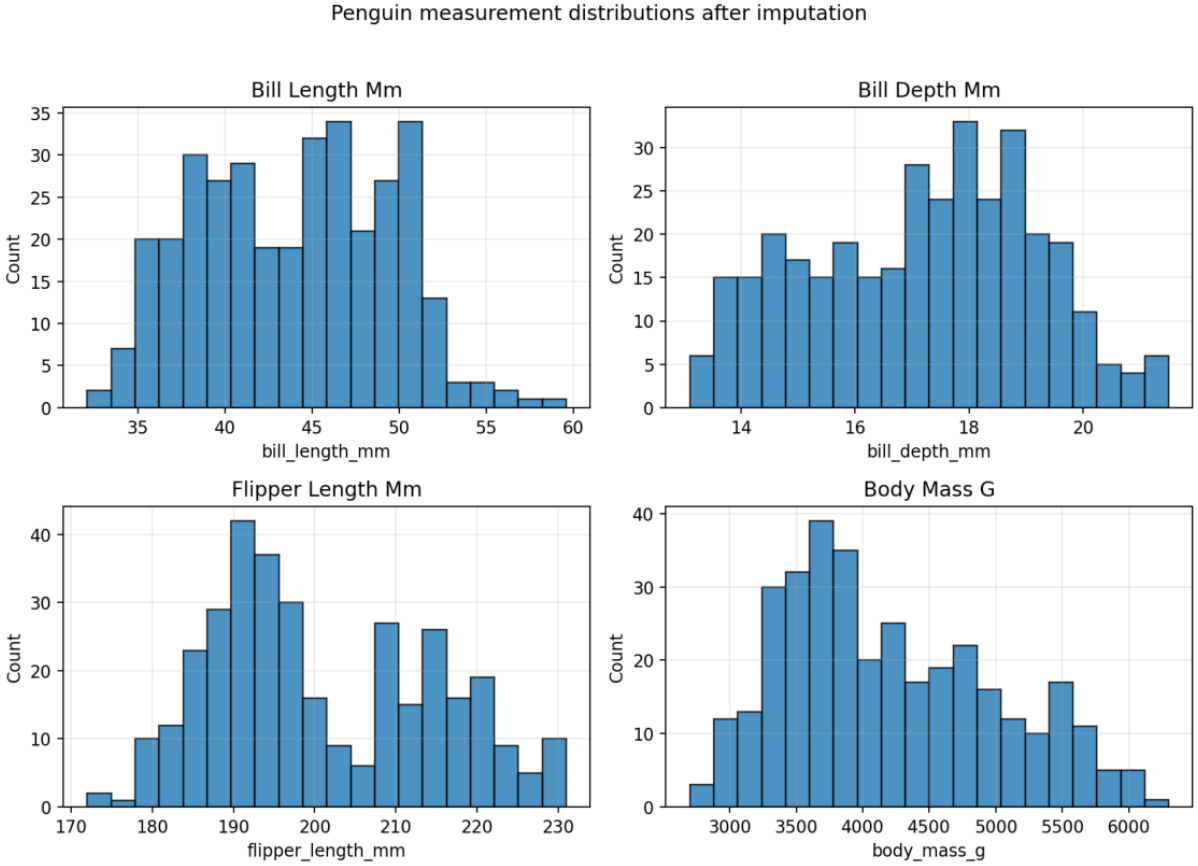


Figure 1. Distributions of the four selected measurements after median imputation.

The two rows with missing selected measurements have missing values in all four selected features. Median imputation retains all 344 observations. Distance-based clustering is sensitive to feature scale: body mass is measured in grams while the bill measurements are in millimetres. Without standardization, the much larger numerical scale of body mass could dominate Euclidean distances and the resulting clusters.

1.1 Exploratory data analysis (continued)

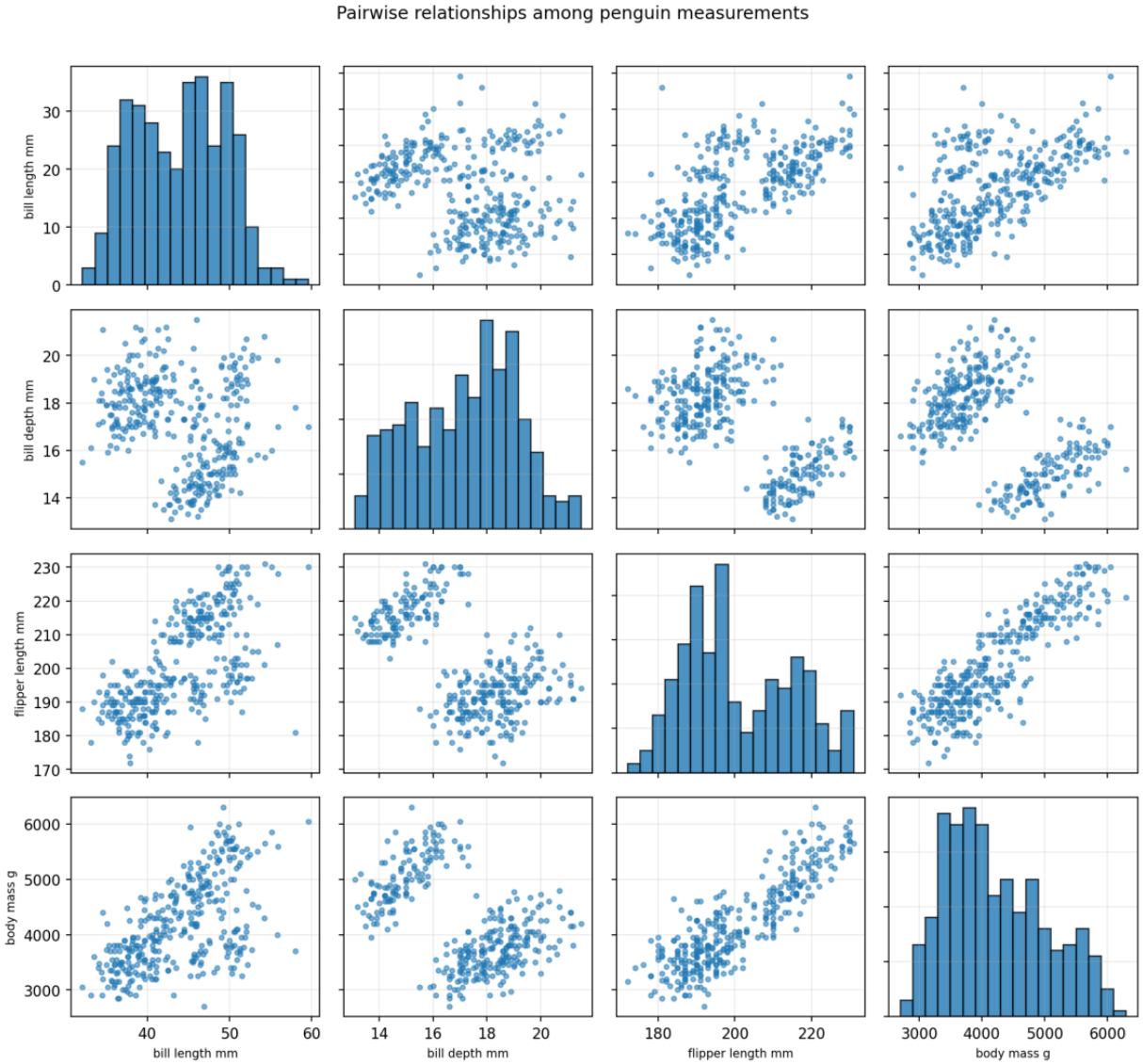


Figure 2. Diagonal panels show measurement distributions; off-diagonal panels show pairwise relationships after imputation and before standardization.

The distributions and pairwise plots show that the measurements have different ranges and non-identical relationships. Standardization transforms each feature to have mean zero and standard deviation one, so a difference of one standard deviation has comparable weight across measurements. The EDA uses only the four numerical measurements; species labels are not used for preprocessing or model selection.

1.2 K-Means clustering

K	Average silhouette score
2	0.5307
3	0.4443
4	0.3976
5	0.3759
6	0.3670

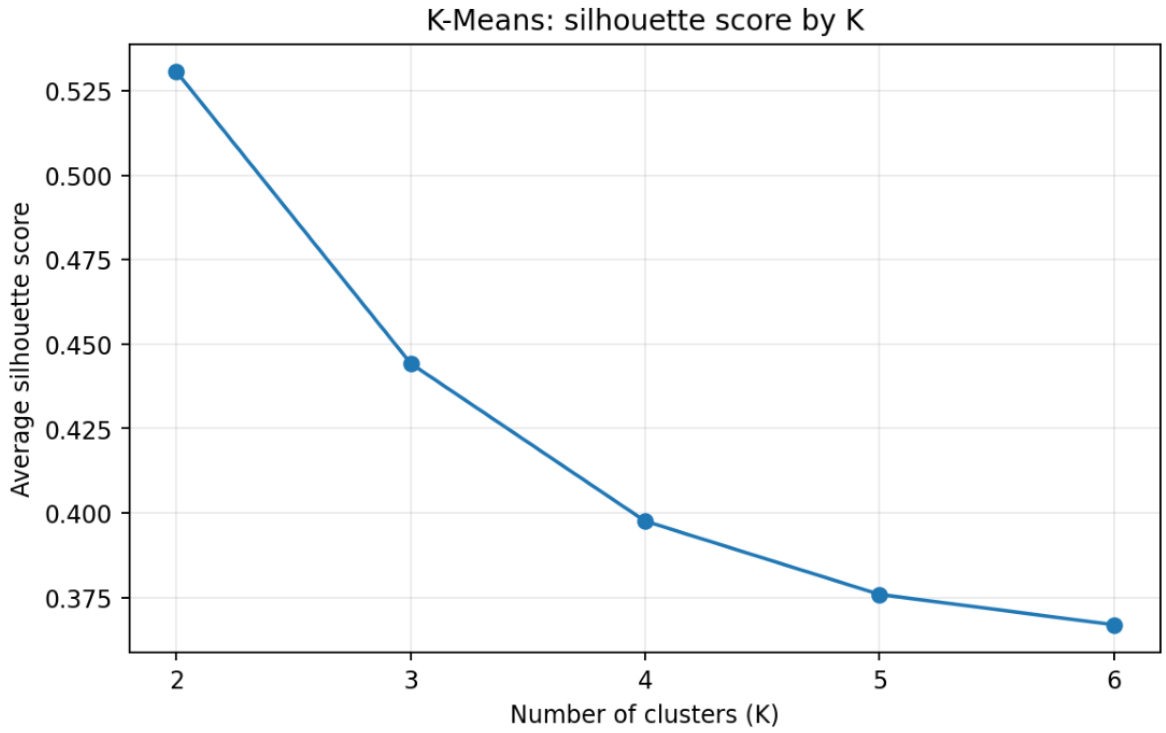


Figure 3. Average silhouette score for K-Means models with K from 2 to 6.

K=2 is selected because it has the highest average silhouette score (0.5307) among the tested values. The silhouette score compares how close each observation is to its own cluster against its nearest alternative cluster, so a larger average value indicates better separation and cohesion for this set of candidate models. Species labels are not used to select K.

1.2 K-Means clustering (continued)

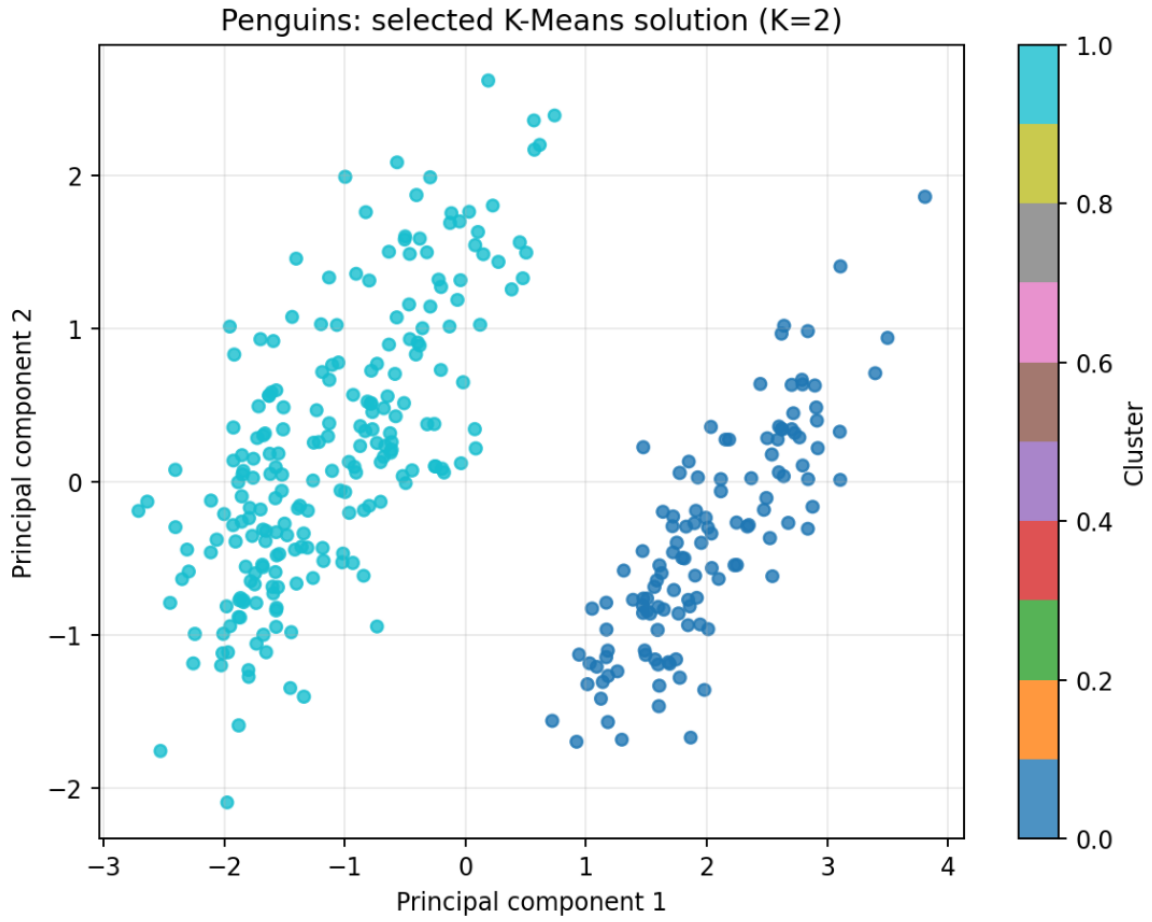


Figure 4. PCA projection of the standardized measurements with selected K-Means assignments (K=2).

Cluster	Observations
Cluster 0	123
Cluster 1	221

A K-Means centroid is the mean feature vector of the observations assigned to a cluster. Its objective can be written as J = sum_i ||x_i - mu_(c_i)||^2, the sum of squared distances from each observation to its assigned centroid. Minimizing this objective encourages compact groups around their means. With Euclidean distance, K-Means naturally favours roughly convex, approximately spherical clusters; it is less suited to elongated, curved, or irregular shapes.

1.3 Gaussian Mixture Models

GMMs were fitted for K=2, 3, 4, 5, 6 and covariance types spherical, diag, tied, and full. The table lists AIC and BIC for every fitted model, sorted by BIC.

K	Covariance type	AIC	BIC
4	tied (selected)	2418.15	2529.53
3	tied	2452.95	2545.12
5	tied	2415.06	2545.64
6	tied	2427.48	2577.26
3	full	2410.24	2579.23
2	full	2498.13	2609.51
4	full	2412.12	2638.72
6	spherical	2538.99	2673.41
2	tied	2616.61	2689.58
5	full	2429.08	2713.29
6	diag	2527.97	2731.52
6	full	2418.32	2760.14
4	spherical	2775.39	2863.73
5	spherical	2774.89	2886.27
3	diag	2802.25	2902.11
4	diag	2767.91	2902.33
5	diag	2756.65	2925.63
3	spherical	2885.81	2951.10
2	diag	2985.29	3050.58
2	spherical	3126.56	3168.81

The lowest-BIC model is GMM with K=4 and tied covariance (BIC=2529.53; AIC=2418.15). AIC and BIC balance fit against model complexity; BIC applies a stronger complexity penalty for this sample size. All 20 candidate fits converged in the validation run.

1.3 Gaussian Mixture Models (continued)

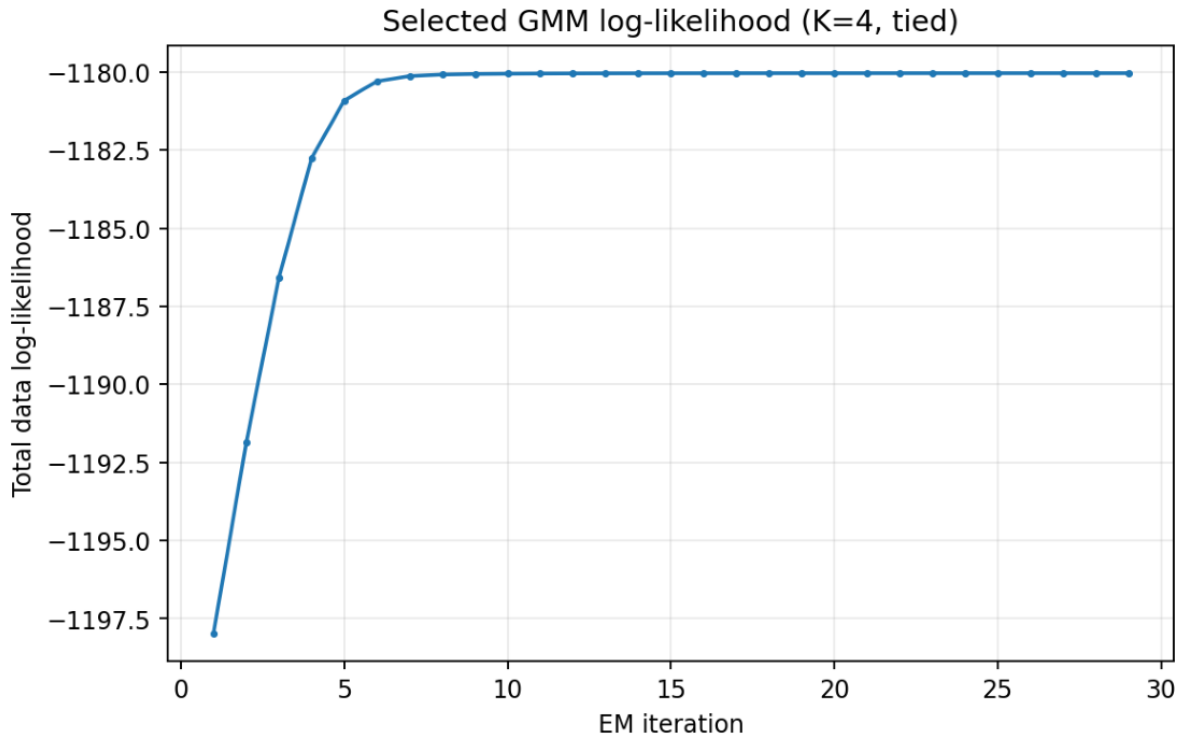


Figure 5. Total data log-likelihood over EM iterations for the selected GMM configuration (K=4, covariance=tied).

The plotted log-likelihood increases from approximately -1197.97 to -1180.03 and levels off. The selected fit converges after 8 EM iterations under the model tolerance; the plotted trace continues with a stricter stopping check to show the objective settling. The GMM mean vector mu_k is the centre of component k. Its covariance matrix Sigma_k describes component spread and feature covariances. Full covariance can represent elongated or rotated components through cross-feature covariance terms, unlike K-Means centroid geometry.

Posterior membership probabilities for five seeded observations

Observation index	P(component 0)	P(component 1)	P(component 2)	P(component 3)
179	1.0000	0.0000	0.0000	0.0000
203	1.0000	0.0000	0.0000	0.0000
213	1.0000	0.0000	0.0000	0.0000
241	1.0000	0.0000	0.0000	0.0000
279	0.0000	0.0153	0.7895	0.1952

K-Means makes a hard assignment to one cluster. A GMM provides soft membership probabilities: values close to 1 indicate strong membership, while comparable probabilities across components indicate uncertainty or overlap. Observation 279 has posterior mass across components 2 and 3, unlike the four rows assigned almost entirely to component 0.

1.4 Comparison with the true species

Ground-truth species labels are used only here, after the K-Means and GMM models have been selected without labels.

Selected model	ARI	NMI
K-Means (K=2)	0.6456	0.7467
GMM (K=4, tied)	0.7484	0.8168

The selected GMM has higher ARI and NMI than the selected K-Means model, so it matches the species partition more closely under these external measures. Silhouette and ARI need not choose the same model: silhouette measures geometric cohesion and separation in the feature space, while ARI and NMI compare cluster assignments with reference labels. The species labels must not guide K selection because doing so would use ground truth during unsupervised model choice. Biological species need not map perfectly to mathematical clusters because measurements overlap, individuals vary, and the four selected measurements may not capture every species distinction.

2. Spatial Clustering with DBSCAN

2.1 Exploring the spatial data

Item	Result
Observations loaded	30,555
Missing latitude values	0
Missing longitude values	0

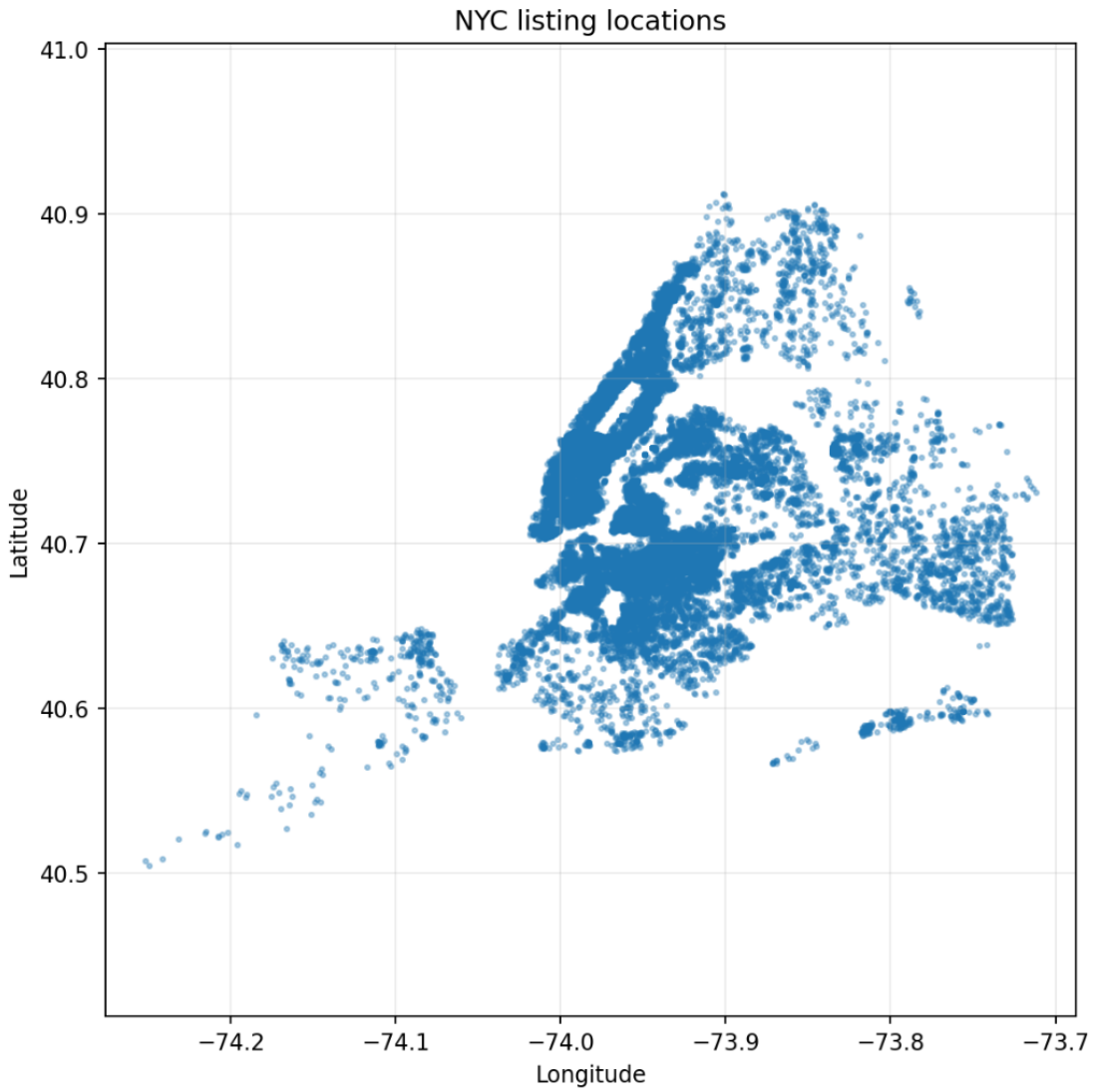


Figure 6. Scatter plot of all available NYC listing coordinates.

The coordinates form dense concentrations alongside more sparsely populated and isolated locations. Spatial activity can follow street networks and neighbourhood boundaries rather than forming approximately circular groups. The later DBSCAN analysis uses great-circle (haversine) distances and reports epsilon in kilometres.

2.2 K-Means on spatial data

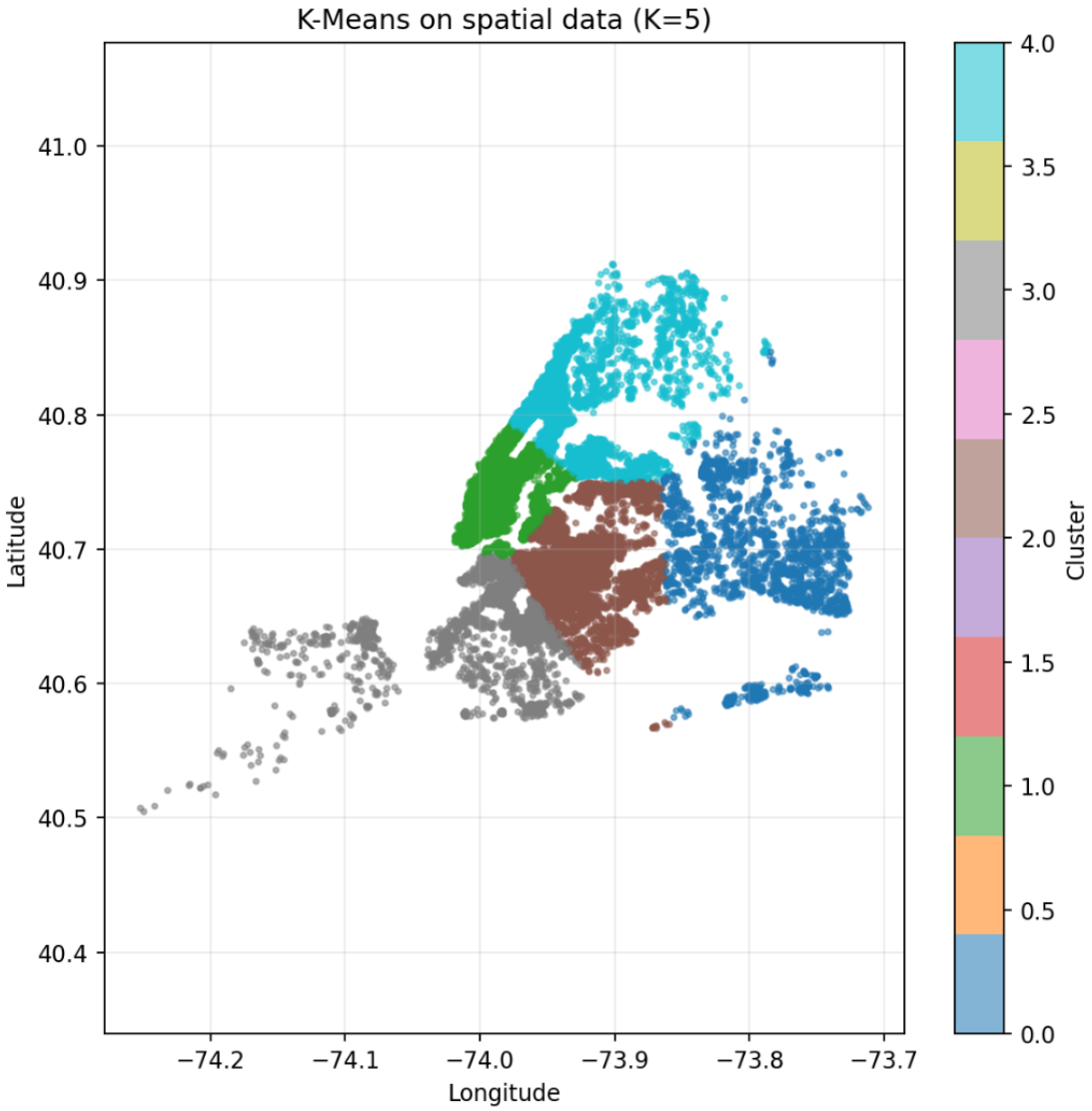


Figure 7. K-Means assignments for the spatial observations (K=5).

Cluster	Observations
Cluster 0	2,155
Cluster 1	11,440
Cluster 2	8,298
Cluster 3	2,971
Cluster 4	5,691

All 30,555 observations are assigned to one of the five clusters. K-Means has no separate noise class, so isolated observations are forced into the nearest-centroid cluster. The partition is based on centroid proximity and may not follow the detailed shape of dense spatial regions.

2.3 DBSCAN and parameter selection (bonus)

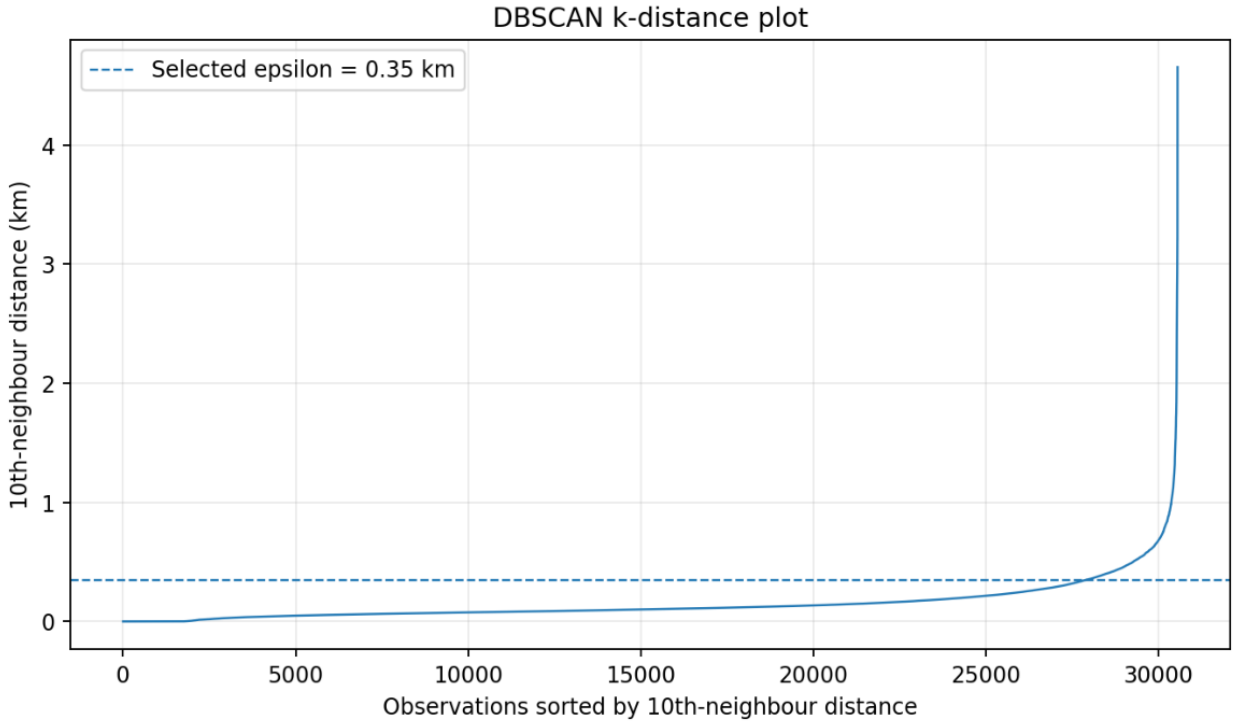


Figure 8. Sorted distance to the 10th nearest neighbour; the dashed line marks the selected starting epsilon.

Parameter / output	Selected result
Distance metric	Haversine (great-circle)
Epsilon	0.35 km
min_samples	10
Discovered clusters	77
Noise points	1,624
Noise observations	5.32%

Epsilon=0.35 km was chosen near the bend in the sorted k-distance curve. `min_samples=10` sets the number of neighbours required for a core point. DBSCAN labels points as noise when they are not density-reachable from a core point under the selected radius and density requirement.

2.3 DBSCAN result (continued)

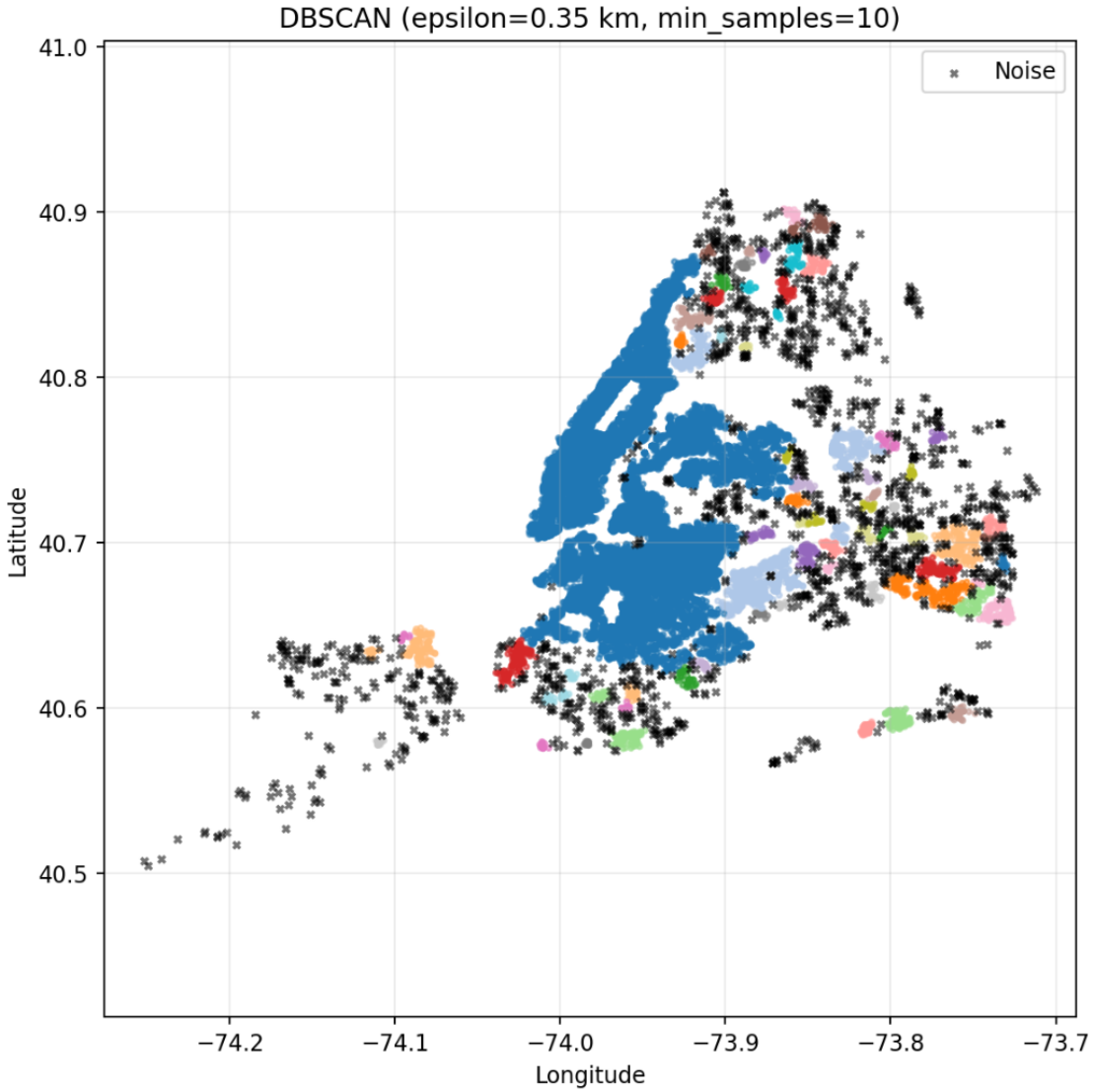


Figure 9. DBSCAN assignments at epsilon=0.35 km and min_samples=10; noise observations use black x markers.

At the selected setting, DBSCAN discovers 77 clusters and labels 1,624 observations (5.32%) as noise. This retains local density-connected groups rather than forcing every listing into a centroid-based partition. The noise markers highlight locations that do not meet the density-reachability conditions for any discovered cluster.

2.4 Understanding the effect of epsilon

Experiment	Epsilon (km)	Clusters	Noise points	Noise (%)
Small epsilon	0.20	128	4,264	13.96%
Selected epsilon	0.35	77	1,624	5.32%
Large epsilon	0.50	34	718	2.35%

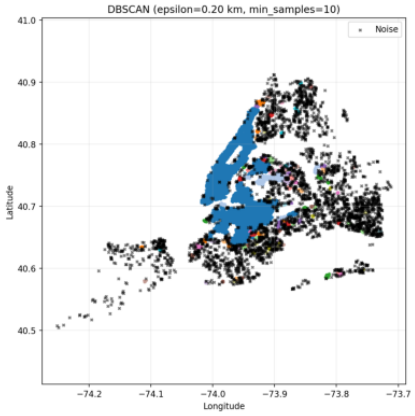


Figure 10. Small epsilon: 0.20 km

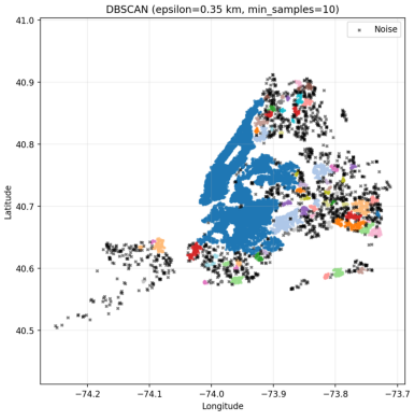


Figure 11. Selected epsilon: 0.35 km

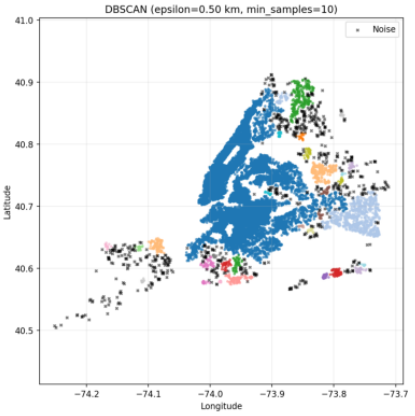


Figure 12. Large epsilon: 0.50 km

As epsilon increases from 0.20 to 0.50 km, the number of clusters falls from 128 to 34 and the noise count falls from 4,264 to 718. A radius that is too small fragments dense areas and leaves more points without enough neighbours. A larger radius connects previously separate neighbourhoods and can merge distinct dense groups.

2.5 K-Means vs. DBSCAN

K-Means assigns every observation to one of K centroid-based clusters, so it has no separate noise label. DBSCAN can label an observation as noise when it lacks support from a sufficiently dense neighbourhood. By connecting density-reachable neighbourhoods, DBSCAN can recover irregularly shaped clusters and does not require K in advance. It is not parameter-free: epsilon and min_samples define the density requirement and affect the numbers of clusters and noise points.

3. Clustering Text Documents

3.1 Text representation

Quantity	Result
Number of documents (N)	2,000
TF-IDF features (D)	8,000
Non-zero entries	114,819
Total matrix entries	16,000,000
Sparsity	99.28%

TF-IDF represents each document by term weights that balance a term's frequency within the document against its frequency across the collection. The matrix is sparse because each document contains only a small subset of the vocabulary. This differs from the penguin data: text uses thousands of mostly-zero term features instead of four dense continuous measurements.

3.2 K-Means on text

K-Means was fitted with K=4 on the original TF-IDF matrix. The ten highest-weight terms in each centroid and an informal topic interpretation are listed below.

Cluster	Top 10 representative terms	Informal topic
Cluster 0	thanks, graphics, mail, image, files, does, hi, gif, program, software	Graphics, images, and software help
Cluster 1	people, just, don, like, think, know, government, make, did, right	General discussion and political opinions
Cluster 2	game, year, baseball, team, games, think, players, hit, runs, don	Baseball and sports
Cluster 3	space, nasa, moon, shuttle, orbit, earth, mission, launch, spacecraft, satellite	Space exploration and astronomy

The terms suggest a graphics/software cluster, a broad general-discussion cluster with political vocabulary, a baseball cluster, and a space/astronomy cluster. These topic names are informal interpretations of centroid terms. A cluster may still be meaningful if its terms form a coherent theme even when it does not reproduce every original newsgroup category exactly, because the model groups documents by textual similarity.

3.3 Dimensionality reduction with Truncated SVD

Quantity	Result
Reduced dimensions	50
Sum of explained variance ratios	0.1199

Representation used with K-Means (K=4)	ARI	NMI
Original TF-IDF	0.3876	0.4958
50-dimensional Truncated SVD	0.2865	0.4498

On this dataset, reducing TF-IDF to 50 dimensions degraded external clustering agreement: ARI decreased from 0.3876 to 0.2865 and NMI decreased from 0.4958 to 0.4498. Dimensionality reduction can help by compressing broad co-occurrence structure into fewer dimensions and reducing computational cost. It can also remove rare but discriminative terms or distinctions between related topics. The 50 components account for a sum of explained variance ratios of 0.1199, so most of the TF-IDF variation is not represented by this reduced space.

3.4 Another clustering model on reduced text (bonus)

A diagonal-covariance GMM with K=4 was fitted to the 50-dimensional SVD representation and compared with K-Means on the same representation. The GMM converged in 47 EM iterations.

Model	Silhouette score in SVD space	Cluster sizes by cluster ID
K-Means on SVD (K=4)	0.0746	212, 310, 379, 1,099
GMM diagonal covariance (K=4)	-0.0341	232, 811, 350, 607

K-Means has the higher silhouette score (0.0746) than the GMM (-0.0341), indicating better geometric separation for this run. The GMM still provides soft probabilistic memberships and component-specific diagonal variances. Its limitations include the simplified covariance assumption and uneven cluster sizes. Text vectors have many dimensions and sparse structure before reduction, so distance and density behaviour differ from two-dimensional geographic data. SVD reduces the calculation size but can discard information needed to distinguish related topics; therefore a method that works well for spatial density clusters may not work equally well on text.

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